

Assignment Operator in SQL Server with Examples

In this article, I am going to discuss **Assignment Operator in SQL Server** with Examples. Please read our previous article, where we discussed [Clauses in SQL Server](#). Before understanding Assignment Operator in SQL Server, let us first understand what are operators and why we need operators, and what are the different types of operators available in SQL Server.

What is an Operator in SQL Server?

An operator is a symbol that performs some specific operation on operands or expressions. These operators are classified as follows in SQL Server.

1. **Assignment operator**
2. **Arithmetic operator**
3. **Comparison operator**
4. **Logical operator**
5. **Set operator**

Note: In this article, I am going to discuss **Assignment Operator**, rest of all other operators will discuss one by one from our upcoming articles.

Understanding the Assignment Operator in SQL Server:

Let us understand how to use the Assignment Operator in SQL Server with an example. We are going to use the following Employee table to understand the Assignment Operator.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
1	Pranaya	Pranaya@g.com	Male	IT	25000	30	Mumbai
2	Tarun	Tarun@g.com	Male	Payroll	30000	27	Odisha
3	Priyanka	Priyanka@g.com	Female	IT	27000	25	Bangalore
4	Preety	Preety@g.com	Female	HR	35000	26	Bangalore
5	Ramesh	Ramesh@g.com	Male	IT	26000	27	Mumbai
6	Pramod	Pramod@g.com	Male	HR	29000	28	Odisha
7	Anurag	Anurag@g.com	Male	Payroll	27000	26	Odisha
8	Hina	Hina@g.com	Female	HR	26000	30	Mumbai
9	Sambit	Sambit@g.com	Male	Payroll	30000	25	Odisha
10	Manoj	Manoj@g.com	Male	HR	30000	28	Odisha
11	Sara	Sara@g.com	Female	Payroll	28000	27	Mumbai
12	Lima	Lima@g.com	Female	HR	30000	30	Bangalore
13	Dipak	Dipak@g.com	Male	Payroll	32000	25	Bangalore

Please use the below script to create and populate the Employee table with the required data.

```
--Create database EmployeeDB
Create database EmployeeDB
Go
Use EmployeeDB
Go
--Create Employee table
CREATE TABLE Employee
(
    ID INT PRIMARY KEY IDENTITY(1,1),
    Name VARCHAR(100),
    EmailID VARCHAR(100),
    Gender VARCHAR(100),
    Department VARCHAR(100),
    Salary INT,
    Age INT,
    CITY VARCHAR(100)
)
GO
--Insert some data into Employee table
INSERT INTO Employee VALUES('Pranaya','Pranaya@g.com','Male','IT', 25000, 30, 'Mumbai')
INSERT INTO Employee VALUES('Tarun','Tarun@g.com','Male','Payroll', 30000, 27, 'Odisha')
INSERT INTO Employee VALUES('Priyanka','Priyanka@g.com','Female','IT', 27000, 25, 'Bangalore')
INSERT INTO Employee VALUES('Preety','Preety@g.com','Female','HR', 35000, 26, 'Bangalore')
INSERT INTO Employee VALUES('Ramesh','Ramesh@g.com','Male','IT', 26000, 27, 'Mumbai')
INSERT INTO Employee VALUES('Pramod','Pramod@g.com','Male','HR', 29000, 28, 'Odisha')
INSERT INTO Employee VALUES('Anurag','Anurag@g.com','Male','Payroll', 27000, 26, 'Odisha')
INSERT INTO Employee VALUES('Hina','Hina@g.com','Female','HR', 26000, 30, 'Mumbai')
INSERT INTO Employee VALUES('Sambit','Sambit@g.com','Male','Payroll', 30000, 27, 'Odisha')
INSERT INTO Employee VALUES('Manoj','Manoj@g.com','Male','HR', 30000, 28, 'Odisha')
INSERT INTO Employee VALUES('Sara','Sara@g.com','Female','Payroll', 28000, 25, 'Bangalore')
INSERT INTO Employee VALUES('Lima','Lima@g.com','Female','HR', 30000, 30, 'Bangalore')
INSERT INTO Employee VALUES('Dipak','Dipak@g.com','Male','Payroll', 32000, 25, 'Bangalore')
GO
```

Assignment Operator:

The assignment operator (=) in SQL Server is used to assign the values to a variable. The equal sign (=) is the only Transact-SQL assignment operator. In the following example, we create the **@MyCounter** variable, and then the assignment operator sets the **@MyCounter** variable to a value i.e. 1.

```
DECLARE @MyCounter INT;
SET @MyCounter = 1;
```

The assignment operator can also be used to establish the relationship between a column heading and the expression that defines the values for that column. The following example displays the column headings as FirstColumn and SecondColumn. The string 'abcd' is displayed for all the rows in the FirstColumn column heading. Then, each Employee ID from the Employee table is listed in the SecondColumn column heading.

```
SELECT FirstColumn = 'abcd', SecondColumn = ID FROM Employee;
```

Compound Assignment Operators in SQL Server:

SQL SERVER 2008 has introduced a new concept of Compound Assignment Operators. The Compound Assignment Operators are available in many other programming languages for quite some time. Compound Assignment Operators are operated where variables are operated upon and assigned in the same line. Compound-assignment operators provide a shorter syntax for assigning the result of an arithmetic or bitwise operator. They perform the operation on the two operands before assigning the result to the first operand.

Example without using Compound Assignment Operators

The following example is without using Compound Assignment Operators.

```
DECLARE @MyVariable INT
SET @MyVariable = 10
SET @MyVariable = @MyVariable * 5
SELECT @MyVariable AS MyResult
GO
```

Example using Compound Assignment Operators

The above example can be rewritten using Compound Assignment Operators as follows.

```
DECLARE @MyVariable INT
SET @MyVariable = 10
SET @MyVariable *= 5
SELECT @MyVariable AS MyResult
GO
```

Following are the list of available compound operators in SQL Server

- +=** Adds some amount to the original value and sets the original value to the result.
- =** Subtracts some amount from the original value and sets the original value to the result.
- *=** Multiplies by an amount and sets the original value to the result.
- /=** Divides by an amount and sets the original value to the result.
- %=** Divides by an amount and sets the original value to the modulo.

In the next article, I am going to discuss [Arithmetic Operators](#) in SQL Server. Here, in this article, I try to explain the **Assignment Operator in SQL Server** with Examples. I hope this article will help you with your needs. I would like to have your feedback. Please post your feedback, question, or comments about this article.